Multiple IV Bolus Practice 4

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

A 192 lb patient is to receive IV bolus injections of a medication every 4 hours. The elimination half-life of the medication is approximately 3.3 hours and the apparent volume of distribution is 30% of body weight.

a Recommend a dose to achieve an average steady state concentration of 25 mg/L. (Round to the nearest 10 mg).

b What is the maximum plasma concentration of the first dose? (The dose is the 550 mg recommended in part a.)

c What is the maximum steady-state plasma concentration? (The dose is the 550 mg recommended in part a.)

d What is the minimum steady-state plasma concentration? (The dose is the 550 mg recommended in part a.)

e What is the plasma concentration 10 hours after the last dose assuming steady state was attained? (The dose is the 550 mg recommended in part a; part c: C_{max}^{∞} = 36.95 mg/L.)

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 550 mg

WHAT THE STEM GIVES

weight	192 lb	$\frac{192}{2.2} = 87.27$ kg, so $V_D = 0.3 \times 87.27 = 26.2$ L
τ	every 4 hours	multiplies V_D and k
t½	3.3 hours	gives $k = \frac{0.693}{3.3} = 0.21\ hr^{-1}$
V_D	30% of body weight	0.3 L per kg × 87.27 kg = 26.2 L
C_{avg}^{∞}	25 mg/L	the target, multiplied by V_D, k and τ
rounding	nearest 10 mg	rounds 550.2 to 550 mg

THE WORKING, ONE LINE AT A TIME

UNIT

$$192\ \frac{lb}{2.2}\ lb/kg = 87.27\ kg;\ V_D = (0.3\ L/kg)(87.27\ kg) = 26.2\ L$$

The weight is given in pounds and the volume as a percentage of body weight, so the weight is converted to kilograms and 0.3 L per kilogram is applied.

SETUP

$$k = \frac{0.693}{3.3\ hr} = 0.21\ hr^{-1}$$

The elimination rate constant is 0.693 over the half-life; reciprocal hours are the unit the exponent and the average need.

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{V_D\ k\ \tau},\ \text{so}\ D_0 = C_{avg}^{\infty}\ V_D\ k\ \tau\ (F = 1)$$

The average steady-state equation is rearranged for the dose: multiplying the target average by the volume, the rate constant and the interval undoes the division. An IV dose is wholly available, so F = 1.

ALGEBRA

$$D_0 = (25\ mg/L)(26.2\ L)(0.21\ hr^{-1})(4\ hr) = 550.2\ mg$$

mg/L × L × hr^{-1} × hr: the litres cancel between mg/L and L, the hours cancel between hr^{-1} and hr, and milligrams are left, the unit of a dose.

ROUND

$$D_0 = 550\ mg$$

The stem asks for the nearest 10 mg, so 550.2 mg is reported as 550 mg; a dose is rounded to an amount that can be measured and given, not left at a tenth of a milligram.

DOES THE NUMBER MAKE SENSE?

Clearance $V_D \times k$ is 26.2 × 0.21 L/hr, a little over 5 L/hr; holding 25 mg/L needs 25 mg for each litre cleared, so well over 100 mg an hour and several hundred milligrams per 4-hour interval. 550 mg fits.

With the unrounded volume, 26.18 L, the dose is 549.8 mg; her key uses 26.2 L and gets 550.2 mg. Both round to 550 mg.

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b

Her key

Answer: 21 mg/L

WHAT THE STEM GIVES

weight	192 lb	$V_D = 0.3 \times \frac{192}{2.2}$ kg = 26.2 L
τ	every 4 hours	*not needed for the first peak*
t½	3.3 hours	*not needed at time zero*
V_D	30% of body weight	26.2 L, the denominator
D_0	550 mg	the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{D_0}{V_D}$$

The first-dose peak is the dose over the volume of distribution, the concentration the moment the bolus is in.

ALGEBRA

$$C_0 = \frac{550 \text{ mg}}{26.2 \text{ L}} = 21 \text{ mg/L}$$

mg over L is mg/L, the concentration the moment the whole 550 mg is in the 26.2 L volume; 550 over 26.2 is 20.99, reported as 21 mg/L.

DOES THE NUMBER MAKE SENSE?

550 mg in 26.2 L: 26.2 × 20 is a little under 550, so a little over 20 mg/L; 21 mg/L fits. The first-dose peak must lie below the steady-state peak.

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c

Her key

Answer: 36.95 mg/L

WHAT THE STEM GIVES

weight	192 lb	$V_D = 0.3 \times \frac{192}{2.2}$ kg = 26.2 L
τ	every 4 hours	the exponent $k\tau = 0.21 \times 4$
t½	3.3 hours	gives $k = \frac{0.693}{3.3} = 0.21$ hr^{-1}
V_D	30% of body weight	26.2 L; $C_0 = \frac{550}{26.2} = 21$ mg/L

D_0	550 mg	$C_0 = \frac{550}{26.2}$ = 21 mg/L, the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max}^{\infty} = \frac{C_0}{1 - e^{-k\tau}}$$

The first-dose peak, 21 mg/L, divided by $(1 - e^{-k\tau})$ adds the drug left over from the earlier doses, so the steady-state peak must come out above 21 mg/L.

ALGEBRA

$$C_{max}^{\infty} = \frac{21 \text{ mg/L}}{1 - e^{-(0.21)(4)}} = \frac{21 \text{ mg/L}}{0.5683} = 36.95 \text{ mg/L}$$

The exponent $(0.21 \text{ hr}^{-1})(4 \text{ hr}) = 0.84$ is a pure number; 4 hours is only 1.2 half-lives, so 43% of each dose remains and the accumulation factor is 1.76.

> **DOES THE NUMBER MAKE SENSE?**
> Four hours is a little over one half-life of 3.3 hours, so a little under half of each dose remains and dividing 21 mg/L by 0.5683 nearly doubles it: about 37 mg/L. The steady-state peak must exceed the 21 mg/L first-dose peak.

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d

Her key

Answer: 15.95 mg/L

WHAT THE STEM GIVES

weight	192 lb	$V_D = 0.3 \times \frac{192}{2.2}$ kg = 26.2 L
τ	every 4 hours	the exponent $k\tau = 0.21 \times 4$
t½	3.3 hours	gives $k = \frac{0.693}{3.3} = 0.21 \text{ hr}^{-1}$
V_D	30% of body weight	26.2 L, the denominator of C_0
D_0	550 mg	C_0 = 21.0 mg/L, the numerator of the trough line

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.30 \times \frac{192 \text{ lb}}{2.2 \text{ lb/kg}} = 26.2 \text{ L}; \; C_0 = \frac{550 \text{ mg}}{26.2 \text{ L}} = 21.0 \text{ mg/L}$$

The volume is 30% of the weight in kilograms, so the pounds are converted first; C_0 is the rise in concentration each 550 mg dose produces.

SETUP

$$k = \frac{0.693}{3.3 \text{ hr}} = 0.21 \text{ hr}^{-1}; \; k\tau = 0.21 \times 4 = 0.84; \; e^{-k\tau} = 0.4317$$

The printed trough line wants k and τ; k comes from the 3.3-hour half-life and the 4-hour interval gives kτ, a pure number.

ALGEBRA

$$C_{min}^{\infty} = \frac{C_0\, e^{-k\tau}}{1 - e^{-k\tau}} = \frac{(21.0\ \text{mg/L})(0.4317)}{1 - 0.4317} = \frac{9.066}{0.5683} = 15.95\ \text{mg/L}$$

This is the printed trough line, read directly with C_0 from the stem: no peak is needed. Part c reaches the same number as the 36.95 mg/L peak times $e^{-k\tau}$.

> **DOES THE NUMBER MAKE SENSE?**
> $k\tau = 0.84$, so 0.4317 of each dose is left when the next goes in; the trough is C_0 times 0.4317 over 0.5683, about three-quarters of the 21.0 mg/L C_0: near 16 mg/L, and below C_0.

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e

Her key

Answer: 4.52 mg/L

WHAT THE STEM GIVES

weight	192 lb	$V_D = 0.3 \times \frac{192}{2.2}$ kg = 26.2 L
τ	every 4 hours	*not needed: the time runs from the last dose*
t½	3.3 hours	gives $k = \frac{0.693}{3.3} = 0.21\ \text{hr}^{-1}$
V_D	30% of body weight	26.2 L; C_0 = 21 mg/L, inside C_{max}^{∞} = 36.95 mg/L from part c
t	10 hours after the last dose	the exponent kt = 0.21 × 10
D_0	550 mg	C_0 = 21 mg/L, inside C_{max}^{∞} = 36.95 mg/L from part c

THE WORKING, ONE LINE AT A TIME

SETUP

$C = C_{max}^{\infty}\, e^{-kt}$, with t = 10 hr

After the last dose at steady state the level falls from the steady-state peak, 36.95 mg/L, by first-order elimination; the 10 hours is counted from the last dose, and the 4-hour interval no longer enters.

ALGEBRA

$$C10\infty = (36.95\ \text{mg/L})\, e^{-(0.21)(10)} = (36.95\ \text{mg/L})(0.1225) = 4.52\ \text{mg/L}$$

The exponent $(0.21\ \text{hr}^{-1})(10\ \text{hr}) = 2.1$ is a pure number; 10 hours is three half-lives, so about one eighth of the peak remains, well below the steady-state trough of 15.95 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> Ten hours is three half-lives of 3.3 hours, so about an eighth (0.125) of the 36.95 mg/L peak remains: near 4.5 mg/L; 4.52 fits and lies below the steady-state trough because 10 hours exceeds the 4-hour interval.
